MIL-C-60441 (MU)
27 April 1967
SUPERSEDING
(See Section 6)

# MILITARY SPECIFICATION

# CONTAINER, SHIPPING AND STORAGE, M473, FOR WARHEAD SECTIONS

## 1. SCOPE

1.1 This Specification cover shipping and storage containers for the following 762MM Warhead Sections:

Warhead Section, HE, M6 and M6E1
Warhead Section, Training, M42

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this Specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | | |
|---|---|---|
| MIL-A-2550 | - | Ammunition and Special Weapons General Specification for |
| MIL-I-45607 | - | Inspection Equipment, Supply and Maintenance |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105) |

FSC: 8140

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| MIL-STD-109 | - | Quality Assurance Terms and Definitions |
| MIL-STD-129 | - | Marking for Shipment and Storage |
| MIL-STD-1168 | - | Lot Numbering of Ammunition |
| MIL-STD-1235 | - | Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes |

DRAWINGS

US ARMY MUNITIONS COMMAND

| | | |
|---|---|---|
| 8813627 | - | Shipping & Storage Container, Warhead Section, M473 |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Container.-The shipping and storage containers shall comply with all requirements specified on Drawing (Dwg) 8813627, and with all requirements specified in applicable specifications.

3.3 Welding.-Welds shall be reasonably smooth and free of craters and pock marks. They shall exhibit characteristics of fusion, penetration and soundness of weld deposit representative of good welding practice. All welding fluxes, scale, weld spatter, acid or basic solutions shall be completely removed prior to application of any finish. The satisfactoriness of the weld shall be based upon compliance with 3.3.1 and 3.3.2.

3.3.1 Qualification.-All welding shall be accomplished in conformance with the process used to quality the first article sample.

3.3.2 Acceptable weld imperfections permitted in each container.

3.3.2.1 Porosity or other discontinuities shall not extend as surface cracks for more than a total of five (5) percent of the total length of each weld.



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3.3.2.2 Blow-outs may be repaired by hand welding operation provided they are no greater than $\frac{1}{4}$ inch in width and one inch in length.

3.4 Assembly.-The container shall be capable of opening and closing freely without evidence of distortion or damage and the simulated warhead (See 6.3), or equivalent assembly, shall assemble and dis-assemble freely when tested as specified in 4.4.1.

3.5 Rough handling.-The container and all accessories shall reveal no significant weakness that would permit damage to the warhead when subjected to tests specified. Minor dents, scratches and deformation that do not obstruct handling, assembly or ability of container to protect contents shall not constitute cause for rejection. The tests shall be performed as specified in 4.4.2.

3.5.1 Handling requirements.-

3.5.1.1 Edgewise drop.-The container shall meet the requirement of 3.5 when tested as specified in 4.4.2.1.

3.5.1.2 Cornerwise drop.-The container shall meet the requirement of 3.5 when tested as specified in 4.4.2.2.

3.5.1.3 Pendulum impact.-The container shall meet the requirement of 3.5 when tested as specified in 4.4.2.3.

3.5.1.4 Roll over.-The container shall meet the requirement of 3.5 when tested as specified in 4.4.2.4.

3.6 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner. They shall be free of burs or sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.7 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract, the specification or as requested by the Contracting Officer. (See 6.4)

3.8 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings, and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "Critical" be accepted solely on the basis of the contractor's recor



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4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of one assembly. All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submmission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-The container will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If the container fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection procedures.

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall



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represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogenity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Changes to the process, specification or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government shall not be deemed to alter the homogeneity of the lot. The inspection lot shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the Contracting Officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period, nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that, inspection for Critical defects shall be 100 percent. Contractors's sampling plans, if used, shall be approved by the Government, and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect except where 100% inspection is specified.

4.3.2.1 Container assembly (See Dwg 8813627).-

| Categories | Defects | Method of | Defect Code (See 6.2) |
|---|---|---|---|
| Critical: | None defined | | |
| Major | AQL 1.00 Percent | | |
| 101. | Component or operation missing | Visual | 01001 |
| 102. | Hardware damaged | Visual | 01002 |
| 103. | Plywood broken | Visual | 01003 |
| 104. | Weld improper | Visual | 01004 |
| 105. | Bolts incorrectly installed | Visual | 01005 |



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| Minor | AQL 1.00 Percent | | |
|---|---|---|---|
| 201. | Marking misleading or unidentifiable | Visual | 01006 |
| 202. | Evidence of poor workmanship (See 3.6) | Visual | 01007 |

4.3.3 Testing.

4.3.3.1 Assembly (See 3.4).-Each container shall be subjected to this test. If any container fails to comply with the applicable requirement, it shall be classified defective and removed from the lot. The test shall be performed as specified in 4.4.1.

4.3.3.2 Rough handling (See 3.5).-The rough handling tests shall be applied to the first article sample only. The sample shall be tested as specified in paragraph 4.4.2.

4.3.4 Government use of contractor's inspection equipment.-The contractor shall, when required by the Government, make available all inspection equipment necessary for determining conformance with contract requirements. Operation of the equipment and verification of its accuracy shall be supplied by the contractor, when required, for the performance of examination or test by the Government.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, constructions, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Assembly, Major Defect, Defect Code (02001).-The container shall be opened, the simulates warhead or equivalent assembly assembled and the container closed. Observation shall be made for evidence of damage during opening or closing of the container or installation of simulated warhead. Subject to approval by the Government the assembly test may be performed in two (2) phases:

a. The assembly of the simulated warhead to the base and suspension assembly.

b. The assembly of the sides, ends and top to the base and suspension assembly. During this portion of the test the locks of the assembly shall be inspected by closing, opening and re-closing on the mating part while the functioning of the lock is checked visually. The locks to be inspected for functioning shall be selected using MIL-STD-105 at an AQL of 0.65%. All of the locks shall be considered equivalent in determining the sample to be inspected.



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4.4.2 Rough handling.-(See 6.3)

4.4.2.1 Edgewise drop, Major Defect, Defect Code (04001).-The container complete with simulated load, shall be supported at one end of its base on a wood sill approximately 5 inches in height, placed normal to the axis of skid and the opposite end will be lifted and dropped from a height of 18 inches onto a concrete or other similarly hard surface. This tests shall be applied to each end of the container. This is a destructive test. Containers so tested shall be discarded.

4.4.2.2 Cornerwise drop, Major Defect, Defect Code (04002).- One corner of the container with simulated load shall be supported by a 5 inch block. A block approximately 12 inches high shall be placed under the other corner of the same end of the container. The lowest corner of the opposite end of the crate shall be raised and allowed to fall freely from a height of 18 inches onto concrete or other similarly hard surface. The test shall be applied to the diagonally opposite corners of the container. This is a destructive test. Containers so tested shall be discarded.

4.4.2.3 Pendulum impact, Major Defect, Defect Code (04003).- The container with simulated warhead shall be freely suspended and shall be swung as a pendulum against a solid, flat and verticle wood, concrete or steel barrier. The container shall be suspended by ropes, chains or cables. The longitudinal axis of the crate shall be perpendicular to the barrier and the end shall rest lightly against it. The container shall be pulled back from the barrier until the center of gravity is raised 18 inches or to the required pendulum angle so that an impact velocity equal to 10 ± 1.5 ft/sec will be attained, and then released and allowed to swing freely against the barrier. This test shall be applied to each end of the container. This is a destructive test. Containers so tested shall be discarded.

4.4.2.4 Roll over, Major Defect, Defect Code (04004).-The container with simulated load, erect on its base on a concrete or other similarly hard surface, shall be tipped slowly sidewise until it falls freely and by its own weight from the base to the side, side to the top, top to the other side, and then to its base, thus completing one revolution. This is a destructive test. Containers so tested shall be discarded.

5. PREPARATION FOR DELIVERY

5.1 Marking.-Marking shall be in accordance with MIL-STD-129.



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5.2 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Simulated warhead.-The simulated warhead section required for these tests shall be furnished by the Government.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.5 Supersession data.-Specification MIL-C-60441 (MU) covering Container, Shipping and Storage M473, supersedes AEI-88 (Rev 2) in part. Container, Shipping and Storage M473A1, will be covered by MIL-C-60442 (MU).

Custodian:
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Preparing activity:
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Project No. 8140-A-147



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

**INSTRUCTIONS**

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

| ORGANIZATION | CITY AND STATE |
|---|---|

| CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT |
|---|---|---|
| | | $ |

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

| SUBMITTED BY *(Printed or typed name and activity)* | DATE |
|---|---|
| | |

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